MODULE
2.4 Practical investigations

2 • Icebergs and eggbergs

Questioning & Predicting | Evaluating

Purpose

To demonstrate how the different densities of fresh and salt water change how something floats.

Hypothesis

Which do you think is more dense—an ice cube or an egg? Before you go any further with this investigation, write a hypothesis in your workbook.

Timing

60 minutes

Materials

PART A: ICEBERGS

- 2 ice cubes
- salt
- water

- 250 mL beaker
- plastic 30 cm ruler
- spoon

PART B: EGGBERGS

- fresh uncooked egg
- salt
- water

- 2 tall glasses, tall beakers or large measuring cylinders, wide enough to take an egg
- spoon

Method

PART A: ICEBERGS

1. Three-quarters fill the glass or beaker with water.
2. Slide in a cube of ice.
3. Use the ruler to measure how much of the ice cube is above water and how much is below water.
4. Use a calculator to calculate the percentage of the ice cube that lies below the water.

$$\% = \frac{\text{how much ice cube is below water}}{\text{total height of ice cube}} \times 100$$

5. Use the spoon to remove the ice cube. Add salt to the water and stir. Keep stirring and adding salt until no more will dissolve.
6. Slide in another cube of ice and repeat your measurements.

PART B: EGGBERGS

7. Fill two tall glasses, tall beakers or large measuring cylinders with approximately the same amount of water.
8. Add spoonfuls of salt to one glass until no more will dissolve.
9. Use the spoon to lower the fresh egg gently into the glass of fresh (non-salty) water. Observe what happens.
10. Lower the egg into the glass of salt water. Observe what happens.
11. Remove the egg and pour out half of the salt water.
12. Very slowly add half of the fresh water, making sure that it does not mix with the salt water. The best way of doing this is by slowly pouring the fresh water down the inside of the glass.
13. Use the spoon to lower the egg gently into the water. Watch where the egg settles.

Results

Record your observations at each stage in Part A and Part B.

Review

1. What evidence suggests that salt water is more dense than fresh water?
2. List the following substances in order from most dense to least dense:

   egg fresh water ice cube salt water
3. Use evidence from this experiment to explain why icebergs are dangerous to shipping.
4. The wreck of RMS *Titanic* was only discovered in 1985. Some scientists think that the wreck may not have hit the bottom immediately but settled somewhere in the water above it. Use observations made in this activity to explain how this might have happened.
5. a Construct a conclusion that compares the densities of ice cubes and eggs.
   b Assess whether your hypothesis was supported or not.
6. An egg that has gone 'off' will float in water whereas a fresh egg will sink. Propose why.